MultiStripMamba: Multi-Scale Strip Convolutions Meet Vision Mamba for Medical Image Segmentation

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Abstract. Many anatomical structures in medical images—such as the kidneys, pancreas, and aorta—exhibit elongated, anisotropic morphologies that are poorly served by the isotropic receptive fields of standard square-kernel convolutions. While strip convolutions have been proposed to address this geometric mismatch [13,44], their potential as a core organizational principle for the decoder in medical image segmentation has not been systematically explored. This paper introduces Multi-Scale Strip Mamba for Medical Image Segmentation (MultiStripMamba), a hybrid architecture that couples a Vision Mamba (VSS) encoder with a decoder and skip connections built around multi-scale strip and large-kernel convolutions. Central to the decoder is the Multi-Scale Convolutional Block (MSCB), which employs parallel strip convolutions at multiple kernel sizes to capture anisotropic features across complementary scales. The Multi-Scale Convolutional Gate (MSCG) extends the same design philosophy to skip connections. Evaluations on Synapse and ACDC benchmarks demonstrate strong segmentation performance, with MultiStripMamba achieving 84.11% DSC and 15.86 mm HD95 on Synapse, and 93.82% DSC on ACDC, while maintaining a compact parameter budget of 11.93M. Ablation studies confirm that strip convolutions are the dominant performance driver, that multi-scale strip kernels substantially outperform their single-scale counterparts, and that the scale spacing—rather than the number—of strip kernels is the critical design factor. Furthermore, the multi-scale MSC alone provides sufficient feature diversity to render additional multi-scale attention modules redundant.

Keywords: Medical Image Segmentation · Vision Mamba · Strip Convolution · Multi-Scale Convolution.

1 Introduction

In abdominal CT, the kidneys extend axially across dozens of slices yet span only a few millimeters laterally. The pancreas curves through the retroperitoneal space with no consistent orientation. The aorta descends as a narrow, elongated tube. These anisotropic morphologies—where an organ's spatial extent varies dramatically across different directions—are the rule rather than the exception in medical imaging, yet they pose a fundamental geometric challenge to the convolutional operations at the heart of modern segmentation architectures. A standard 3×3 or 5×5 convolution is isotropic by design: it aggregates features uniformly in all spatial directions, making no distinction between the elongated contour of a kidney and a uniform background region. This geometric mismatch between isotropic convolution kernels and anisotropic anatomy is the central problem this work addresses.

Effective segmentation of such structures demands two complementary capabilities. First, the model must maintain a sufficiently broad receptive field to disambiguate organ boundaries from surrounding anatomy—a requirement traditionally met by self-attention mechanisms in Transformers [4,20,25], but at the cost of quadratic computational complexity. Vision Mamba [6,7,8], built upon selective state space models, has recently emerged as an efficient alternative, offering global context modeling with linear complexity. Initial adaptations to medical segmentation, such as VM-UNet [9] and U-Mamba [10], construct U-shaped architectures where both encoder and decoder are composed of VSS blocks. However, while the Mamba encoder effectively captures global dependencies, the VSS-block decoder lacks the convolutional inductive biases essential for fine-grained spatial detail recovery, motivating a hybrid design where the Mamba encoder is paired with an enhanced convolutional decoder.

The second requirement—and the primary focus of this work—is a decoder capable of capturing the directional, anisotropic features that characterize real anatomical structures. Strip convolutions [13,44,45], which decompose a square kernel into orthogonal 1D strip kernels ($1 \times N$ and $N \times 1$), offer a principled geometric solution: they concentrate the receptive field along a specific direction, matching the elongated morphology of organs far more naturally than isotropic square kernels. However, a single strip kernel size cannot simultaneously match the diverse anisotropy scales present across different organs—from the finely elongated pancreas to the broadly extended aorta. We hypothesize that the systematic use of multiple strip kernel sizes, and critically, the specific choice of which sizes to combine, is key to unlocking the full potential of strip convolutions in medical segmentation.

To test this hypothesis, we propose MultiStripMamba, a hybrid architecture built around multi-scale strip convolutions as the core organizational principle of the decoder. The encoder is a standard hierarchical VSS network, following VM-UNet [9], providing efficient global context modeling. The decoder introduces the Multi-Scale Convolutional Block (MSCB), a three-stage pipeline (CA→SA→MSC) in which Channel Attention and Spatial Attention first re-calibrate the feature maps, after which the Multi-scale Convolution (MSC) module applies parallel strip convolution branches at distinct kernel sizes to capture anisotropic features across complementary scales. The same design philosophy extends to skip connections through the Multi-Scale Convolutional Gate (MSCG), where strip convolutions operate alongside standard convolutions to produce directionally-aware gating signals.

The principal contributions of this work are threefold:

1. We identify the geometric mismatch between isotropic convolutions and anisotropic organ morphology as a critical and under-explored issue in medical segmentation decoder design, and propose multi-scale strip convolutions as the structurally appropriate inductive bias for this domain.
2. We design MultiStripMamba, in which the MSCB decoder and MSCG skip connections place multi-scale strip convolutions at the center of a hybrid Mamba-CNN architecture, replacing the conventional VSS-block decoder.
3. Experiments on Synapse and ACDC demonstrate strong segmentation accuracy (84.11% / 93.82% DSC) with 11.93M parameters. Ablation reveals that (i) strip convolutions are the dominant performance driver, (ii) multi-scale strip kernels substantially outperform single-scale variants, (iii) the scale spacing between kernels—not their quantity—is the governing design principle, and (iv) the multi-scale MSC alone provides sufficient feature diversity to render additional multi-scale attention modules redundant.

2 Related Works

This section reviews related work in medical image segmentation and state space models, providing the context for our proposed approach.

2.1 Medical image segmentation

The U-Net [1] and its encoder-decoder variants have long served as the foundational architecture for medical image segmentation. Numerous extensions have been proposed, including attention-based skip connections [34,35], dense skip pathways [36], and Transformer-based encoders and bottlenecks [4,20,25]. Our work builds on this lineage, focusing specifically on the decoder and skip connections rather than the encoder.

Of particular relevance is the line of work on strip-shaped operations for capturing anisotropic features. Strip Pooling [13] introduced elongated pooling kernels to capture long-range spatial context along one dimension. SLaK [44] demonstrated that large-kernel convolutions can be efficiently factorized into strip components. Active Strip Convolution [45] proposed learnable, direction-adaptive strip kernels. These methods collectively establish that strip-shaped operations are geometrically better suited to elongated structures than square kernels. However, their application has largely been confined to backbone networks or general-purpose modules; their systematic integration as the central organizing principle of a medical segmentation decoder—particularly at multiple complementary scales—remains unexplored.

Attention gates, first introduced for medical segmentation by Attention U-Net [34], suppress irrelevant features in skip connections while amplifying salient ones. Our MSCG extends this concept by incorporating strip convolutions into the gating mechanism, producing attention coefficients that are sensitive to both spatial context and directional structure.

2.2 State Space Model

State Space Models (SSMs), deeply rooted in classical control theory and signal processing, provide a robust mathematical framework for modeling dynamical systems [28]. These models characterize the evolution of a system's latent states through linear ordinary differential equations (ODEs) or their discrete-time counterparts, offering a principled methodology for processing both continuous and discrete sequences [30]. Historically, SSMs have been recognized for their rigorous theoretical foundations and their innate capacity to capture long-range dependencies with linear computational complexity—a property that stands in stark contrast to the quadratic scaling inherent in the self-attention mechanism of Transformers [6,24]. By effectively mapping input sequences to hidden representations through a recursive or convolutional formulation, SSMs have emerged as a highly efficient alternative for modeling extensive temporal or spatial contexts [29,31].

The application of State Space Models (SSMs) to deep learning accelerated significantly with the introduction of the Structured State Space Sequence model (S4) [30]. S4 integrated the HiPPO (Hidden Information Processing for Preserving Online Trajectories) initialization framework for long-range memory with a structured, linear time-invariant (LTI) formulation, facilitating highly efficient parallel training and demonstrating remarkable performance on extensive sequence modeling tasks [29]. However, a critical limitation of standard S4 and similar LTI-SSMs is their intrinsic time-invariant property—their state-transition parameters remain fixed and strictly independent of the input data [31]. This structural rigidity fundamentally restricts their capacity to perform context-dependent reasoning or to selectively focus on salient information within a complex sequence, a critical requirement for tasks involving dense and heterogeneous visual representations [6].

To overcome these constraints, the selective state space model, notably Mamba [6], was introduced. The core innovation of Mamba lies in parameterizing the SSM transition matrices as functions of the input, thereby incorporating a data-dependent selection mechanism. This allows the model to dynamically propagate or attenuate information based on the current context, significantly enhancing its representation capacity for information-dense data such as natural language and medical imagery [8,7]. By maintaining the computational efficiency of linear scaling while introducing this selective capability, Mamba-based architectures have emerged as a high-performance alternative to Transformers, effectively bridging the gap between global receptive fields and computational economy in complex vision tasks [9,10].

3 Methodology

This section first introduces the overall architecture of MultiStripMamba, followed by detailed descriptions of each constituent module.

3.1 MultiStripMamba

Figure 1 (a) presents the overall architecture of MultiStripMamba, which follows the U-shaped encoder-decoder design [1]. Specifically, MultiStripMamba comprises a Patch Embedding layer for initial tokenization, four-stage VSS Blocks [7,6] as the encoder, four-stage MSCBs as the decoder, a Segmentation Head (SH) layer for final projection, and MSCGs as skip connections.

In the Patch Embedding layer, the input image $x \in \mathbb{R}^{H \times W \times 3}$ is partitioned into non-overlapping 4×4 patches, producing $x' \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}$ via linear projection to dimension C. Before entering the encoder, x' is normalized via layer normalization.

For the encoder, eight VSS Blocks are distributed across four stages, with each stage comprising two VSS Blocks. A patch merging operation is applied at the end of the first three stages to downsample the spatial dimensions of the input features while simultaneously doubling the channel capacity [32]. Consequently, the channel counts for the four stages are denoted as $[C, 2C, 4C, 8C]$.

For the decoder, four Multi-Scale Convolutional Blocks (MSCBs) progressively refine the hierarchical pyramid features extracted from the encoder via skip connections (i.e., x_1, x_2, x_3, and x_4, as illustrated in Figure 1) [33]. A patch expanding operation is employed following each of the first three MSCBs to progressively restore the spatial resolution [20]. The channel counts for the decoder stages follow the reverse order: $[8C, 4C, 2C, C]$.

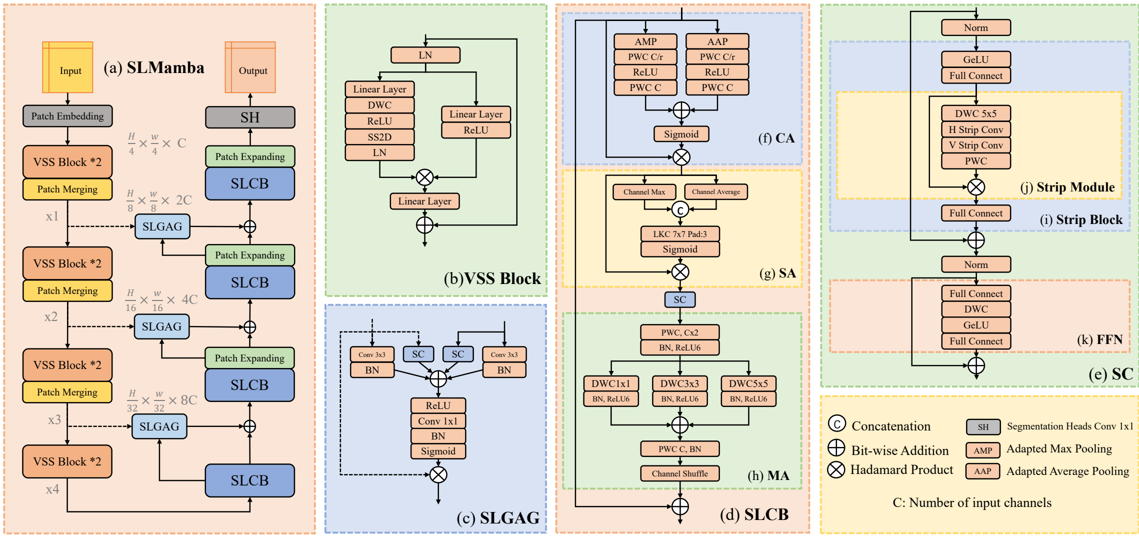

Fig. 1. (a) The overall architecture of MultiStripMamba. (b) VSS Block is the main construction of our encoder. (c) Multi-Scale Convolutional Gate (MSCG) can enhance features from skip connections. (d) Multi-Scale Convolutional Block(MSCB), progressing patches in all dimensions to further enhance features, is the main component of our decoder. (e) Multi-scale Convolution (MSC). (f) Channel attention block(CA). (g) Spatial attention block(SA). (h) Strip block. (i) Strip module. (j) Feed-Forward Network(FFN).

To bridge the semantic gap between the encoder and decoder, three Multi-Scale Convolutional Gates (MSCGs) are integrated within the skip connections [34]. The features that are upsampled in the patch expanding layer are subsequently element-wise added to the refined outputs from the corresponding MSCGs.

Finally, a Segmentation Head (SH) layer is applied to project the fused features and produce the final segmentation output.

3.2 VSS Block

As depicted in Figure 1 (b), the VSS Block, derived from VMamba [8,7], serves as the core module of our encoder. The input patches x are first layer-normalized and then split into two parallel branches: one branch performs global context modeling, while the other preserves local features.

The global branch processes the input through a linear layer $LL(\cdot)$, a depthwise separable convolution $DWC(\cdot)$, and an activation function $R(\cdot)$. The resulting features are then fed into $SS2D(\cdot)$ for selective scan-based feature extraction, followed by layer normalization $LN(\cdot)$. The local branch applies only a linear layer $LL(\cdot)$ and an activation function $R(\cdot)$, retaining fine-grained spatial details.

The outputs from both branches are merged via a Hadamard product after each passes through an additional linear layer $LL(\cdot)$. Finally, the merged features are added to the initial input x via a residual connection.

The $VSS(.)$ can be formulated as in Equation 1:

$$\begin{aligned} x_g &= \mathrm{LL}(x) \\ g &= \mathrm{LL}\big(SS2D(\mathrm{LN}(\mathrm{DWC}(R(x_g))))\big) \\ l &= \mathrm{LL}(R(x_g)) \\ VSS(x) &= x + g \otimes l \end{aligned} \tag{1}$$

3.3 Multi-Scale Convolutional Gate (MSCG)

We introduce the Multi-Scale Convolutional Gate (MSCG) to enhance the activation of relevant features while suppressing irrelevant ones.

As shown in Figure 1 (c), MSCG accepts two inputs: g from a patch expanding layer in the decoder, and x from the corresponding encoder stage via skip connection. Both inputs share identical spatial dimensions to enable valid element-wise operations. Each input is processed by a 3×3 convolution ($Cnv_{3\times3}(\cdot)$) and a strip convolution ($MSC(\cdot)$) in parallel. The resulting feature maps are batch-normalized ($BN(\cdot)$) and merged via element-wise addition. The combined feature map is activated by a ReLU function $R(\cdot)$, then passed through a 1×1 convolution ($Cnv_{1\times1}(\cdot)$) to produce a single-channel attention map. This attention map is subsequently normalized by $BN(\cdot)$ and passed through a sigmoid activation $\sigma(\cdot)$ to yield the attention coefficients.

Finally, the initial skip-connection features x are element-wise multiplied by the attention coefficients to produce the gated output.

The $MSCG(.)$ can be formulated as in Equation 2:

$$\begin{aligned} g' &= MSC(g) + BN(Cnv_{3\times3}(g)) \\ x' &= MSC(x) + BN(Cnv_{3\times3}(x)) \\ \alpha &= \sigma(BN(Cnv_{1\times1}(R(g' + x')))) \\ MSCG(g, x) &= x \otimes \alpha \end{aligned} \tag{2}$$

3.4 Multi-Scale Convolutional Block (MSCB)

We introduce the Multi-Scale Convolutional Block (MSCB) as the core building block of our decoder, as depicted in Figure 1 (d). The MSCB comprises three sub-modules arranged sequentially. A Channel Attention block (CA) and a Spatial Attention block (SA) first re-calibrate the feature maps along the channel and spatial dimensions, respectively. The re-calibrated features then enter the Multi-scale Convolution (MSC)—the central component of the MSCB—which employs multiple parallel strip convolutions at different kernel sizes to capture anisotropic structures across complementary scales. This three-stage design (CA→SA→MSC) prioritizes the multi-scale strip convolution pathway while keeping the attention overhead minimal.

Channel Attention Block (CA)[37]: As depicted in Figure 1 (f), the input x first enters the CA module and is split into two parallel branches. In each branch, x undergoes adaptive max pooling ($AMP(\cdot)$) and adaptive average pooling ($AAP(\cdot)$), respectively [38], to extract the most salient per-channel statistics from the entire feature map. For each pooled descriptor, two point-wise convolution blocks [39] are applied: $PWC_{C/r}(\cdot)$ (with reduction ratio $r = 16$) to compress the channel dimension, and $PWC_C(\cdot)$ to restore it, with an activation function $R(\cdot)$ placed between them. The two resulting feature vectors are combined via element-wise addition and passed through a sigmoid activation $\sigma(\cdot)$. Finally, the resulting channel attention weights are element-wise multiplied with the initial input x.

The $CA(.)$ can be formulated as in Equation 3:

$$\begin{aligned} a_{amp} &= PWC_C(R(PWC_{C/r}(AMP(x)))) \\ a_{aap} &= PWC_C(R(PWC_{C/r}(AAP(x)))) \\ CA(x) &= x \otimes \sigma(a_{amp} + a_{aap}) \end{aligned} \tag{3}$$

Spatial Attention Block (SA)[37]: As depicted in Figure 1 (g), following a design analogous to the CA module, the input x is first processed along the channel dimension: max pooling ($CH_{max}(\cdot)$) and average pooling ($CH_{average}(\cdot)$) are applied to aggregate spatial information into two 2D descriptors. These descriptors are concatenated and then fed into a large-kernel convolution $LKC(\cdot)$ with a 7×7 kernel (padding $= 3$) to capture local spatial context. The resulting feature map is passed through a sigmoid activation $\sigma(\cdot)$ and element-wise multiplied with the initial input x.

The $SA(.)$ can be formulated as in Equation 4:

$$SA(x) = x \otimes \sigma(LKC([CH_{max}(x); CH_{average}(x)])) \tag{4}$$

3.5 Multi-scale Convolution (MSC)

The Multi-scale Convolution (MSC) [43,44,45] is the central component of the MSCB, designed to address the limitations of standard square-kernel convolutions in capturing elongated and curvilinear anatomical structures. While prior work has employed a single strip kernel size to model anisotropic features, a fixed kernel size cannot optimally match the diverse scales of anisotropy present across different organs—from the finely elongated pancreas to the broadly extended aorta.

To address this, we extend the strip convolution to a multi-scale formulation. Rather than using a single pair of strip kernels ($1 \times N$ and $N \times 1$), the MSC module employs multiple strip kernel pairs in parallel, each with a distinct N, to capture anisotropic features at complementary scales. As illustrated in Figure 1 (e), the input x first passes through layer normalization, a fully connected layer, and a GELU activation function [46]. The normalized features x' then enter the Strip Module, the core component of MSC, as depicted in Figure 1 (h).

Within the Strip Module, a large-kernel depthwise convolution (5×5) first extracts local contextual features. Subsequently, multiple parallel branches of depthwise strip convolutions are applied. Each branch consists of a horizontal strip kernel ($1 \times N_i$) and a vertical strip kernel ($N_i \times 1$), where different N_i values target different degrees of structural anisotropy. The outputs from all branches are summed, followed by a point-wise convolution for channel mixing.

The output of the Strip Module, denoted as Z, undergoes a Hadamard product with x' to produce Z', which is then element-wise added to a fully-connected projection of x. This intermediate result is added to x once more, yielding Y. Finally, Y is normalized and processed by a feed-forward network (FFN), with the FFN output added to the normalized Y via a residual connection to produce the final output $MSC(x)$.

The $MSC(.)$ can be formulated as in Equation 5:

$$\begin{aligned} x' &= \mathrm{GELU}(\mathrm{FC}(\mathrm{LN}(x))) \\ Z &= \mathrm{StripModule}(x') \\ MSC(x) &= x + \mathrm{FFN}\big(\mathrm{LN}\big(x + \mathrm{FC}(x) + x' \otimes Z\big)\big) \end{aligned} \tag{5}$$

where the StripModule captures multi-scale anisotropic features through a large-kernel depthwise convolution followed by parallel strip convolution branches:

$$\mathrm{StripModule}(z) = \mathrm{PWC}\Big(\mathrm{DWC}_{5\times5}(z) + \sum_i \big(\mathrm{DWC}_{1\times N_i}(z) + \mathrm{DWC}_{N_i\times1}(z)\big)\Big) \tag{6}$$

where $\mathrm{DWC}_k(\cdot)$ denotes a depthwise convolution with kernel size k, $\mathrm{PWC}(\cdot)$ denotes a point-wise convolution for channel mixing, and the kernel sizes are $N_i \in \{7, 11\}$ in our final configuration.

3.6 Loss Function

To optimize the MultiStripMamba for high-precision medical image segmentation, we employ a hybrid loss function, denoted as L_{total}, which synergistically combines Soft Dice Loss and Focal Loss.

The rationale behind this design is two-fold. First, Soft Dice Loss is utilized to maximize the spatial overlap between the segmented masks and the ground truth, effectively mitigating the issues caused by the severe class imbalance inherent in medical datasets (e.g., small lesion areas versus large background regions). Second, to address the challenge of hard-to-classify pixels near anatomical boundaries, Focal Loss is integrated. By introducing a modulating factor $(1 - p_t)^\gamma$, Focal Loss reshapes the standard cross-entropy loss to down-weight easy examples and focus the model's learning capacity on ambiguous regions and sparse foreground pixels.

The objective function is formulated as in Equation 7:

$$L_{total} = \lambda_{dice} L_{Dice} + \lambda_{focal} L_{Focal} \tag{7}$$

where λ_{dice} and λ_{focal} are hyper-parameters representing the weighting coefficients for each component. In our experiments, we empirically set $\lambda_{dice} = 1$ and $\lambda_{focal} = 1$ to maintain a balanced gradient flow throughout the backpropagation process, ensuring both global structural consistency and local boundary sharpness.

4 Experiments

This section will present the details of implementation followed by a comparative analysis of our MultiStrip-Mamba against SOTA methods, including the performance of the complete model on the datasets and the details of the ablation study.

4.1 Implementation details

To evaluate the proposed MultiStripMamba, comprehensive experiments are conducted on two widely adopted public benchmarks: the Synapse multi-organ segmentation dataset [47] and the Automated Cardiac Diagnosis Challenge (ACDC) dataset [48].

Datasets. The Synapse dataset comprises 30 abdominal CT scans with 13 abdominal organs annotated. Following the standard protocol [4,5], we adopt the official split of 18 training cases and 12 testing cases. The ACDC dataset contains 100 cardiac MRI scans from 100 patients, each with annotations for the right ventricle (RV), left ventricle (LV), and myocardium (Myo). We follow the standard split of 70 training cases, 10 validation cases, and 20 testing cases [48]. Patient-level separation is strictly enforced to prevent data leakage between training and test sets.

Preprocessing. All images are uniformly resized to a spatial resolution of 224×224, and intensity values are normalized to zero mean and unit variance. To mitigate the risk of overfitting and enhance generalization, standard data augmentation techniques including random flipping and random rotation are employed.

Training Protocol. We set the batch size to 32 and employ the AdamW optimizer [49] with an initial learning rate of 1×10^{-5} and weight decay of 1×10^{-3}. A cosine annealing learning rate scheduler [50] is applied with a maximum of 50 iterations and a minimum learning rate of 1×10^{-5}. The entire training process spans 300 epochs. The random seed is set to 42. For the hybrid loss function, the Focal Loss parameters are set to $\alpha = 0.5$ and $\gamma = 0.5$, and the Dice Loss uses a smoothing factor of $\epsilon = 1 \times 10^{-3}$.

Evaluation. For quantitative assessment, two primary metrics are used: the Dice Similarity Coefficient (DSC) to measure region-based overlap [51], and the 95% Hausdorff Distance (HD95) to evaluate boundary-based accuracy [52].

All experiments are implemented using the PyTorch framework and executed on a single NVIDIA RTX 5090 GPU.

4.2 Experimental Results

We compare MultiStripMamba against a range of state-of-the-art methods on both Synapse and ACDC datasets. The baselines cover multiple architectural families: CNN-based models such as UNet[1] and AttnUNet[34], Transformer-based models such as Swin-UNet[5,20] and MISSFormer[61], Mamba-based models such as VM-UNet[9], and hybrid architectures such as TransUNet[4] and TransCASCADE[53]. For fair comparison, all baseline results are reproduced under our experimental setting unless otherwise noted. The results demonstrate that MultiStripMamba consistently achieves the best performance across both datasets.

Results on Synapse Multi-Organ Segmentation Results on Synapse are presented in Table 1. Among the compared methods, MultiStripMamba achieves the best overall performance with a mean DSC of 84.11% and a mean HD95 of 15.86 mm, outperforming TransCASCADE[53] (82.68% / 17.34 mm) and VM-UNet[9] (81.08% / 19.21 mm). The improvement stems from two factors. First, the Vision Mamba encoder provides efficient global context modeling with linear complexity. Second, and more critically, the multi-scale MSC within the MSCB decoder captures anisotropic structures across complementary scales, providing stronger local feature refinement than standard decoders. [TODO: per-organ analysis]

Figure 2 presents qualitative comparisons on the Synapse dataset. Several baseline models exhibit notable segmentation failures: TransUNet[4], Swin-UNet[5], and VM-UNet[9] all fail to adequately segment the left kidney. Additionally, TransUNet and Swin-UNet miss a substantial portion of the pancreas. In contrast, MultiStripMamba produces segmentation masks that more faithfully delineate the ground truth, particularly for small and slender organs such as the kidneys and pancreas.

Results of ACDC Segmentation Results on ACDC are presented in Table 2. MultiStripMamba achieves the best overall performance with a mean DSC of 93.82%, surpassing TransCASCADE[53] (92.12%) and VM-UNet[9] (91.63%). [TODO: per-structure analysis]

Figure 3 presents qualitative comparisons on the ACDC dataset. Swin-UNet[5] incorrectly omits the middle portion of the left ventricle, while TransUNet[4] misses the left portion of the myocardium. Although VM-UNet[9] and TransCASCADE[53] produce approximately correct segmentations, their boundary delineation is less smooth and precise compared to MultiStripMamba, which yields the most anatomically faithful contours.

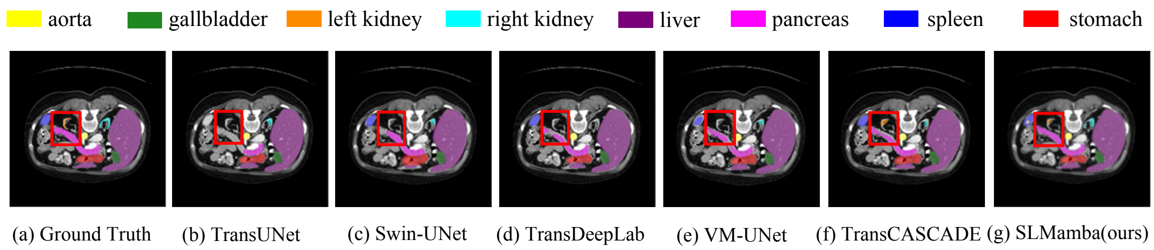

Fig. 2. Qualitative results on Synapse datasets are shown. The red rectangular box highlights incorrectly segmented structures by SOTA methods.

Methods	DICE↑	HD95↓	Aor.	Gal.	Kid.(L)	Kid.(R)	Liv.	Pan.	Spl.	Sto.
nnU-Net[3]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
UNet[1]	70.11	44.69	84.00	56.70	72.41	62.64	86.98	48.73	81.48	67.96
AttnUNet[34]	71.70	34.47	82.61	61.94	76.07	70.42	87.54	46.70	80.67	67.66
TransUNet[4]	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
Swin-UNet[5]	79.56	21.55	85.47	66.53	83.28	79.61	94.29	56.58	90.66	76.60
UNETR[25]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
PVT-CASCADE[58]	81.06	20.23	83.01	70.59	82.23	80.37	94.08	64.43	90.1	83.69
Swin-UMamba[11]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
VM-UNet[9]	81.08	19.21	86.40	69.41	86.16	82.76	94.17	58.80	89.51	81.40
TransCASCADE[53]	82.68	17.34	86.63	68.48	87.66	84.56	**94.43**	**65.33**	90.79	83.52
MultiStripMamba(ours)	**84.11**	**15.86**	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]

Table 1. Results on Synapse Multi-Organ Segmentation. DICE scores are reported for individual organs. ↑ denotes higher the better, ↓ denotes lower the better. The best results are in bold.

4.3 Ablation Studies

Effect of Individual MSCB Components To isolate the contribution of each sub-module within the MSCB, we conduct a systematic ablation study by removing one component at a time from the full model. Specifically, we evaluate the following variants: (1) without Channel Attention ($-$CA), (2) without Spatial Attention ($-$SA), and (3) without Multi-scale Convolution ($-$MSC). All experiments are conducted on the Synapse dataset under identical training settings.

[TODO: analysis of component ablation results]

Effect of Multi-scale Convolution A fixed strip kernel size cannot simultaneously match the diverse anisotropy scales of different anatomical structures. To investigate this, we extend the MSC module from a single strip kernel pair to multiple parallel pairs with distinct kernel sizes, and evaluate all seven combinations on the Synapse dataset. All experiments use the same VSS encoder and MSCG to isolate the effect of the MSC module.

Several findings emerge. First, multi-scale strip convolution consistently outperforms any single-scale variant: $[7,11]$ achieves 84.19% DSC, a substantial +0.83 improvement over the single-scale $[11]$ baseline. Second, and most critically, the scale spacing between strip kernels—not the kernel size or quantity—is the governing design principle. The $[7,11]$ configuration (26.21M parameters) outperforms $[9,11]$ (32.93M parameters, 83.89% DSC) despite having fewer parameters. We attribute this to the wider scale gap between 7 and 11 (spanning fine local anisotropy to broad elongated structures) compared to the narrow gap between 9 and 11, which produces largely overlapping receptive fields. The absence of a small kernel (e.g., 7) in $[9,11]$ also means that fine-grained directional textures near organ boundaries are lost. Third, adding a third kernel ($[7,9,11]$) yields only a marginal further gain (+0.34 DSC) at the cost of nearly doubling the parameter count (47.60M), confirming diminishing returns from incremental scale coverage. We therefore adopt $[7,11]$ as the final MSC configuration.

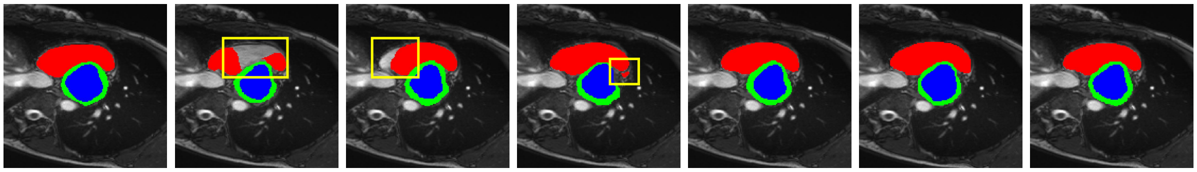

Fig. 3. Qualitative results on ACDC datasets are shown. Red represents the right ventricle, blue represents the left ventricle, and green represents the myocardium. The yellow rectangular box highlights incorrectly segmented regions by SOTA methods.

Methods	DICE↑	RV	Myo	LV
nnU-Net[3]	[TODO]	[TODO]	[TODO]	[TODO]
UNet[1]	[TODO]	[TODO]	[TODO]	[TODO]
AttnUNet[34]	[TODO]	[TODO]	[TODO]	[TODO]
TransUNet[4]	89.71	86.67	87.27	95.18
SwinUNet[5]	90.00	88.55	85.62	95.83
HiFormer[60]	90.12	91.06	84.54	94.77
MISSFormer[61]	90.86	89.55	88.04	94.99
VM-UNet[9]	91.63	90.25	89.14	95.50
TransCASCADE[53]	92.12	90.65	89.68	96.02
MultiStripMamba(ours)	**93.82**	[TODO]	[TODO]	[TODO]

Table 2. Results on ACDC. DICE scores(%)are reported for individual organs.↑ denotes higher the better. The best results are in bold.

Effect of Decoder Architecture To isolate the contribution of the proposed MSCB decoder, we fix the VSS encoder and compare three decoder variants on both Synapse and ACDC datasets. **VSS decoder**: the decoder is constructed using VSS blocks, following the design of VM-UNet [9], where each decoder stage mirrors the encoder's VSS block structure. **Standard CNN decoder**: each decoder stage consists of two consecutive 3×3 convolutions with batch normalization and ReLU activation, following the original U-Net decoder design [1]. **MSCB decoder (ours)**: the full proposed decoder with CA, SA, and MSC sub-modules. All three variants share the identical VSS encoder, patch expanding layers, and skip connections (without MSCG, to isolate the decoder effect). Table 4 reports the results. [TODO: analysis of results]

Effect of MSCG To evaluate the effectiveness of the MSCG mechanism, we conduct an experiment in which the MSCG-enhanced skip connections are replaced with standard unprocessed skip connections, and compare this variant against the full MultiStripMamba on both Synapse and ACDC datasets. The results are reported in Table 7. [TODO: analysis of MSCG results]

Effect of Loss Function To evaluate the contribution of each loss term, we compare three training configurations on the Synapse dataset: (1) Soft Dice Loss only, (2) Focal Loss only, and (3) the combined Dice + Focal Loss (our default). The weighting coefficients and hyperparameters follow the settings described in Section 3.6.

5 Conclusions

This paper has presented MultiStripMamba, a medical image segmentation framework that pairs Vision Mamba-based global encoding with multi-scale strip convolutional decoding, built on the hypothesis that strip convolutions should be the central organizing principle of the decoder. By integrating the VSS Block encoder with the proposed MSCB decoder—centered on multi-scale strip convolutions—and MSCG skip connections, the model effectively balances global context modeling with fine-grained local feature extraction,

Methods	Params (M)↓	FLOPs (G)↓	Inference (ms)↓	GPU Mem. (GB)↓	Train Time (h)↓
UNet[1]	34.53M	65.53G	[TODO]	[TODO]	[TODO]
Swin-UNet[5]	27.17M	5.95G	[TODO]	0.31 GB	[TODO]
TransUNet[4]	105.28M	593.00G	[TODO]	7.55 GB	[TODO]
VM-UNet[9]	27.43M	3.15G	[TODO]	0.42 GB	[TODO]
TransCASCADE[53]	35.12M	8.29G	[TODO]	[TODO]	[TODO]
MultiStripMamba(ours)	11.93M	2.27G	[TODO]	[TODO]	[TODO]

Table 3. Computational efficiency comparison. Params: number of parameters; FLOPs: floating-point operations per input (measured at 224×224); Inference: average inference time per image; GPU Mem.: peak GPU memory during inference; Train Time: total training time on Synapse. ↓ denotes lower is better.

Variant	Synapse		ACDC
	DICE↑	HD95↓	DICE↑
VSS decoder	[TODO]	[TODO]	[TODO]
Standard CNN decoder	[TODO]	[TODO]	[TODO]
MSCB decoder (ours)	**84.11**	**15.86**	**93.82**

Table 4. Ablation study of decoder architectures on Synapse and ACDC datasets. All variants use the same VSS encoder. ↑ denotes higher the better, ↓ denotes lower the better.

achieving 84.11% DSC and 15.86 mm HD95 on Synapse, and 93.82% DSC on ACDC, with a compact 11.93M parameter budget.

Ablation studies yield three key findings. First, strip convolutions are the dominant performance driver within the decoder. Second, multi-scale strip kernels substantially outperform single-scale variants, and the scale spacing between kernels—rather than their quantity—is the governing design principle. Third, the multi-scale strip convolution alone provides sufficient feature diversity, rendering additional multi-scale attention modules redundant and enabling a simpler, more parameter-efficient decoder. We hope these findings encourage further exploration of geometrically-motivated convolutional designs in medical image segmentation.

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Variant	Synapse DSC↑	HD95↓
MultiStripMamba (full)	84.11	15.86
− CA	[TODO]	[TODO]
− SA	[TODO]	[TODO]
− MSC	[TODO]	[TODO]

Table 5. Ablation study of individual MSCB components on the Synapse dataset. Each variant removes one component while keeping all others unchanged.

MMSC Kernel Sizes	Params (M)	FLOPs (G)	Synapse DSC↑
[7]	[TODO]	[TODO]	[TODO]
[9]	[TODO]	[TODO]	[TODO]
[11] (single-scale baseline)	17.20	2.63	83.36
$[7, 9]$	[TODO]	[TODO]	[TODO]
$[9, 11]$	32.93	5.19	83.89
$[7, 11]$ (ours)	**26.21**	**4.18**	**84.19**
$[7, 9, 11]$	47.60	7.74	84.53

Table 6. Effect of multi-scale convolution kernel configurations on the Synapse dataset. All experiments include MA ($[1, 3, 5]$) to isolate the MSC contribution. In the final architecture, MA is removed (see text), yielding the reported 84.11% DSC with 11.93M parameters.

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Methods	Synapse			ACDC
	DICE↑	HD95↓	Kid.(L)	DICE↑
MultiStripMamba W/o MSCG	[TODO]	[TODO]	[TODO]	[TODO]
MultiStripMamba	**84.11**	**15.86**	[TODO]	**93.82**

Table 7. Results of Ablation Experiment of how Multi-Scale Convolutional Gate (MSCG) affects on Synapse and ACDC datasets. ↑ denotes higher the better, ↓ denotes lower the better.

Loss Configuration	Synapse DSC↑	HD95↓
Dice Loss only	[TODO]	[TODO]
Focal Loss only	[TODO]	[TODO]
Dice + Focal Loss (ours)	**84.11**	**15.86**

Table 8. Ablation study of loss function configurations on the Synapse dataset.

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